LaTeX

PENG

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1 TikZ

Definition 1.1 (TikZ)**.** *TikZ is a frontend of PGF.*

2 Curved Path Construction

Definition 2.1 (control points)**.**

```
(start_point) .. controls (control_point_1) and (control_point_2) .. (end_point)
```

```
\begin{tikzpicture}[scale=2]
    \filldraw [gray]
    (0,0) circle (2pt)
    (1,1) circle (2pt)
    (2,1) circle (2pt)
    (2,0) circle (2pt);
    \draw [thick] (0,0) .. controls (1,1) and (2,1) .. (2,0);
    \draw [-latex] (0,0) -- node[sloped,above] {tangent} (1,1);
    \draw [-latex] (2,0) -- node[sloped,below] {tangent} (2,1);
\end{tikzpicture}
```



A Tools

Example A.1 (magic comment)**.**

```
% !TEX program = lualatex
\usepackage{fontspec}
\setmonofont{DejaVu Sans Mono}[Scale=MatchLowercase]
```

Example A.2 (tikz to png)**.**

```
#!/bin/bash

# convert TikZ PDF to PNG
# pdf2png PDF PAGE PNG [DPI]
pdf2png() {
    local PDF=$1
    local PAGE=$2
    local PNG=$3
    local DPI=${4:-300}
    echo "Convert $PDF page $PAGE to $PNG"
    magick -density "$DPI" -units pixelsperinch \
        "$PDF[$PAGE]" \
        -alpha remove -background white -flatten \
        -trim +repage \
        -quality 100 \
        "$PNG"
}
```

Example A.3 (compress pdf)**.**

```
#!/bin/bash

# compress PDF
# pdfcompress INPUT OUTPUT
pdfcompress() {
    local INPUT=$1
    local OUTPUT=$2
    ghostscript -sDEVICE=pdfwrite -dPDFSETTINGS=/prepress -dNOPAUSE -dQUIET -dBATCH \
        -sOutputFile="$OUTPUT" "$INPUT"
}
```